

Professional Summary

Data Scientist and AI/ML specialist with 5+ years of experience architecting and deploying advanced machine learning solutions, particularly in predictive modeling for Electric Vehicles (EVs) and energy systems. Proven expertise in deep learning, time-series forecasting (Autoformer, Informer, TFT), and MLOps within automotive and utility-relevant contexts, consistently driving innovation to improve predictive accuracy and operational efficiency. Adept at translating cutting-edge research into production-ready applications and eager to apply these skills to develop robust EV predictors for the utilities sector.

Career Objectives

- Develop and deploy advanced predictive models for Electric Vehicle (EV) behavior and energy consumption within the utilities sector
- Leverage expertise in time-series forecasting and machine learning to optimize EV integration with energy grids
- Translate data-driven insights into actionable strategies for utility companies navigating the transition to e-mobility
- Contribute to innovative solutions at the intersection of EV technology, data science, and energy management

Core Competencies

Machine Learning: Expert in predictive modeling, deep learning, and advanced time-series forecasting (Autoformer, Informer, TFT, Hugging Face Transformers), with applications in Electric Vehicle (EV) analytics and energy systems.

MLOps: Proficient with CI/CD pipelines (Airflow, Tekton, AWS CodeDeploy), load balancing (AWS), EC2 Auto Scaling, model deployment using Nginx & Gunicorn, and regulatory-compliant workflows.

Data Engineering: Hands-on experience with AWS (S3, EC2, CodeDeploy, Vertex AI, Dataproc), GCP, Docker, microservices, REST APIs, WebSocket integration, SQL, MongoDB, and ETL pipelines.

Software Development: Proficient in Python, Java, SwiftUI, and modern software engineering best practices. Experience using Optuna and Hugging Face Trainer API for hyperparameter optimization.

NLP & Computer Vision: Skilled in NLP (Hugging Face) and computer vision for real-time analytics & remote monitoring applications.

Code Analysis & Optimization: Developed an intelligent agent for automated code complexity analysis, identifying performance bottlenecks and recommending actionable refactoring strategies to enhance software efficiency and maintainability.

Communication & Leadership: Excellent technical writing, mentorship, and presentation skills for multidisciplinary teams and stakeholder engagement.

Professional Experience

Ford Motor Company
Data Scientist

Detroit, MI, USA
Aug 2023 – Present

- Spearheaded the design and implementation of a robust data ingestion and processing pipeline for large-scale EV charging session data, enabling advanced analytics and predictive modeling for charging behavior and energy demand
- Built transformer-based time-series models (Autoformer, Informer, TFT) to improve parking probability predictions by **10%**, leveraging Hugging Face Transformers for modular experimentation
- Deployed scalable microservices with Nginx & Gunicorn on AWS, processing over **100K data points/sec** and reducing overall latency by **25%** through optimized load balancing
- Employed Optuna & Hugging Face Trainer API for advanced hyperparameter tuning, cutting model training time by **40%** with multi-GPU and parallelized workflows
- Integrated Google Cloud Vertex AI and Dataproc for large-scale distributed model training, ensuring seamless data ingestion and compliance with enterprise security standards
- Implemented secure data pipelines and real-time dashboards, enabling business stakeholders to access production analytics in a compliant environment

Key Projects:

- **Predictive Modeling for EV Energy Bill Reduction:** Developed an XGBoost regression model to predict percentage-based energy bill savings for EV users by optimizing Vehicle-to-Home/Grid (V2H/V2G) participation; engineered features from vehicle telematics, utility rate structures, parking behavior, and home energy consumption, achieving an **R-squared of 0.85**
- **ML-Driven EV Discharge Optimization:** Designed machine learning components to optimize EV battery discharge schedules for V2H/V2G services; analyzed discharge opportunity ratios, home energy load, and parking confidence to predict optimal discharge (kWh)

CereVu Medical, Inc.

San Francisco Bay Area, CA

Software Engineer & Computer Research Scientist

Jun 2023 – Sep 2024

- Architected an end-to-end SwiftUI-based iOS app for real-time biosignal monitoring, utilizing WebSocket for ultra-low-latency data transfer and AWS backend services
- Led MLOps pipeline optimization (AWS CodeDeploy, Tekton), reducing deployment time by **30%** with automated rollbacks and EC2 Auto Scaling for global traffic bursts
- Designed robust data infrastructure with MongoDB databases and integrated load balancing on AWS, enabling secure and scalable data exchange between iOS apps & cloud services
- Built and deployed AI algorithms (via Nginx & Gunicorn) into wearable medical patches, boosting real-time patient monitoring accuracy by **15%**
- Enhanced data processing throughput by **20%** using advanced concurrency patterns and microservices, accelerating biosignal interpretation and response times

Applied Cognitive Ergonomics Lab, Texas A&M University

College Station, TX

Research Assistant

Sep 2021 – May 2024

- Engineered deep learning pipelines for wearable-based stress detection, elevating accuracy by **76%** through personalized feature extraction and advanced sensor fusion
- Developed *GlucoseCoach* iOS app for real-time, personalized glucose monitoring, demonstrating significant improvements in patient adherence and health outcomes
- Incorporated hybrid ML models and explainable AI, enhancing clinician trust and interpretability in high-stakes healthcare decisions
- Leveraged computer vision algorithms for COVID-19 compliance monitoring, achieving **98%** accuracy in detecting face mask usage and social distancing

Department of Psychological & Brain Sciences, Texas A&M University

College Station, TX

Research Assistant

Sep 2018 – Aug 2021

- Created *OpenSync* to integrate EEG/ERP and biosensor data via WebSocket pipelines, enabling high-resolution, real-time behavioral analysis
- Applied advanced signal processing to EEG datasets, improving data fidelity and uncovering critical insights into cognitive processes
- Developed protocols unifying multiple biosensors for enriched cognitive workload assessments, accelerating research throughput across projects
- Implemented AI-driven gait analysis (Conv-LSTM), reaching **93%** accuracy in identifying gait anomalies and potential fall risks

Education

Texas A&M University

College Station, TX

Ph.D. in Industrial Engineering (Data Science)

Aug 2021 – Dec 2023

Dissertation: *"Advancing Stress Detection: Real-Time, Naturalistic, and Personalized AI Approaches for Stress and Mental Disorder Detection"*

GPA: 4.0/4.0 *Summa Cum Laude*

Texas A&M University

College Station, TX

M.S. in Computer Science

Aug 2019 – May 2021

Relevant Coursework: Machine Learning, Deep Learning, NLP, Algorithms, Pattern Recognition, Computer Vision, Database Systems, Reinforcement Learning, Data Mining

GPA: 4.0/4.0 *Summa Cum Laude*

Certifications

- **Advanced MLOps Specialization** (Coursera) – End-to-end MLOps best practices with real-world applications
- **NLP Specialization** (Coursera) – Advanced NLP techniques & large-scale language modeling
- **Deep Learning Specialization** (DeepLearning.AI) – Fundamentals & advanced neural network architectures
- **Bayesian Statistics** (Coursera) – Probabilistic modeling & inference for data-driven decisions
- **Generative AI with LLMs** (AWS) – Practical, hands-on generative AI methodologies

Selected Publications

2024: Razavi, M., et al. *Integrated Eye-Tracking and EEG Data Collection and Synchronization for VR-Based Spatial Ability Assessments.* *IHSI*

2023: Razavi, M., et al. *Machine Learning, Deep Learning, and Data Preprocessing for Stress Detection: A Scoping Review.* *JMIR*

2023: Janfaza, V., Weston, K., Razavi, M., et al. *MERCURY: Accelerating DNN Training via Input Similarity.* *IEEE HPCA*

2023: Kim, N., Grégoire, L., Razavi, M., et al. *Virtual Accident Curb Risk Habituation in Workers.* *iScience*

2022: Razavi, M., et al. *Automatic Monitoring of Physical Distancing & Mask Wearing.* *SN Computer Science*

2021: Razavi, M., et al. *OpenSync: Synchronizing Multiple Measures in Neuroscience.* *Journal of Neuroscience Methods*